

AJAY VENKATESH
Brooklyn, NY
+1 516 585 9013 | in | |

Skills

Programming Languages:

Web Technologies:

Database:

Cloud:

Tools:

AI/ML:

Experience

Campus Mesh

Jan 2026 – Present

Software Engineer

New York, US

Developed and maintained scalable Node.js microservices powering real-time academic workflows for 10K+ users, including study groups, messaging, notifications, and profile management.

Built a personalized recommendation engine using collaborative filtering and behavioral analytics, increasing user engagement by 28% and retention by 50%.

Worked on LLM-based retrieval pipelines (RAG) for CampusIQ (AI Assistant), leveraging embedding-based semantic search and structured data indexing to improve contextual accuracy by 30%.

Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the system design for approval and established Role Based Access (4+ roles) using OAuth, Midway SSO, and HELIS.

Provisioned serverless Infrastructure-as-Code using AWS CDK with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by 70% and enabling multi-stage, multi-region CI/CD deployments.

Built a React 18 frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by 40%.

Engineered backend services in Java using Coral, Smithy, Dagger, and CBOR serialization; applied low-level design and design patterns, reduced backend execution time by 35% and ensured by unit and integration testing using Mockito.

New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer (On-Campus)

New York, US

Architected an NSF-funded (\$100K) AI-powered health app for iOS and Android, processing real-time wearable data; deployed cloud-hosted AI models for health metrics analysis and food image processing.

Engineered ETL pipelines transforming NoSQL into PostgreSQL using concurrency control and locking mechanisms; processed 250K+ records and powered AWS QuickSight dashboards for real-time health analytics.

Developed an insurance analytics platform (React, Node.js, REST APIs); deployed Docker containers on EC2, implemented Stripe billing, RBAC, and row-level security, generating \$50K+ revenue.

Streamlined underwriting via broker portal integrated with insurer platform; implemented document and media processing and WebSocket-based real-time communication; secured 6 insurance clients.

PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Developed a distributed backend on Microsoft Azure for a health platform, integrating with Microsoft CRM and implementing event-driven processing and fault tolerance.

Led code reviews and mentored engineers on software design principles, reducing production bugs by 25%.

Deployed an NLP based OCR neural model on Azure to automate structured data extraction, reducing data processing time by 75%.

Optimized data ingestion with parallel processing for 100K-row from Excel; reduced execution time from 5 hours to 1 hour.

Built Power BI dashboards processing large datasets from multiple databases, improving reporting efficiency by 40%.

Projects

Advanced Expense Tracker (Java Desktop Application) | Java, JavaFX, JDBC, SQLite, JFreeChart

Architected a modular Java application supporting individual and group expenses with secure login, applying OOP principles and MVC & DAO design patterns.

Implemented concurrent transaction handling and real-time cost splitting using multithreading and synchronized blocks to ensure thread-safe database updates.

Hospital Management System (Full-Stack Application) | React, Node.js, Express, MongoDB, SageMaker, Hugging Face

Built a full-stack healthcare platform using React, Node.js, Express, and MongoDB to manage appointments, patient records, lab reports, and medical history with role-based access control.

Fine-tuned a Hugging Face transformer-based model achieving 88% accuracy for health condition prediction and deployed it on AWS SageMaker for real-time diagnostic support in the doctor dashboard.

Education

New York University | Master of Science in Computer Engineering (GPA – 3.9/4)

Sep 2024 – May 2026

Certifications

AWS Cloud Practitioner

Azure Data Fundamentals

Microsoft Solution Architect Expert

Power BI Data Analyst Associate

Microsoft Developer Associate

Stanford Machine Learning (Coursera)